

Supplementary Material for "TabSurv: Tabular Foundation Model for Breast Cancer Prognosis using Gene Expression Data"

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1 Pathway Enrichment Analysis

To assess the biological coherence of the feature set used by TabSurv, we performed a pathway enrichment analysis on the 94 DRIVER_CORE genes. Of these, 90 genes were successfully mapped to Entrez gene identifiers using the org.Hs.eg.db annotation database [3] and used for subsequent analysis. The enrichment of Gene Ontology (GO) [1] was carried out in terms of the biological process, and the enrichment of the KEGG [4] pathway was carried out, with adjusted p-values using the Benjamini-Hochberg method [2] (adjusted $p < 0.05$ considered significant).

1.1 Gene Ontology Enrichment

GO Biological Process analysis identified 1,453 significantly enriched terms (adjusted $p < 0.05$). The most enriched processes were gland development (26 genes; adjusted $p = 6.02 \times 10^{-18}$), regulation of the cell cycle phase transition (26 genes; adjusted $p = 7.39 \times 10^{-18}$) and the G1/S phase transition of the cell cycle (22 genes; adjusted $p = 8.46 \times 10^{-18}$). Additional terms highly enriched included epithelial cell proliferation (25 genes; adjusted $p = 1.75 \times 10^{-16}$), radiation response (24 genes; adjusted $p = 3.41 \times 10^{-16}$), and regulation of epithelial cell proliferation (24 genes; adjusted $p = 1.06 \times 10^{-16}$). Phosphatidylinositol 3-kinase signalling was significantly enriched (14 genes; adjusted $p = 1.09 \times 10^{-12}$), consistent with the inclusion of components of the PI3K pathway in the DRIVER_CORE set. Cellular senescence (12 genes; adjusted $p = 1.71 \times 10^{-11}$) and negative regulation of the cell cycle (20 genes; adjusted $p = 6.44 \times 10^{-13}$) were also among the terms most enriched, reflecting the representation of tumour suppressor mechanisms. The top 15 enriched GO terms are shown in Figure S1a, and the complete list is available in SM-GO.

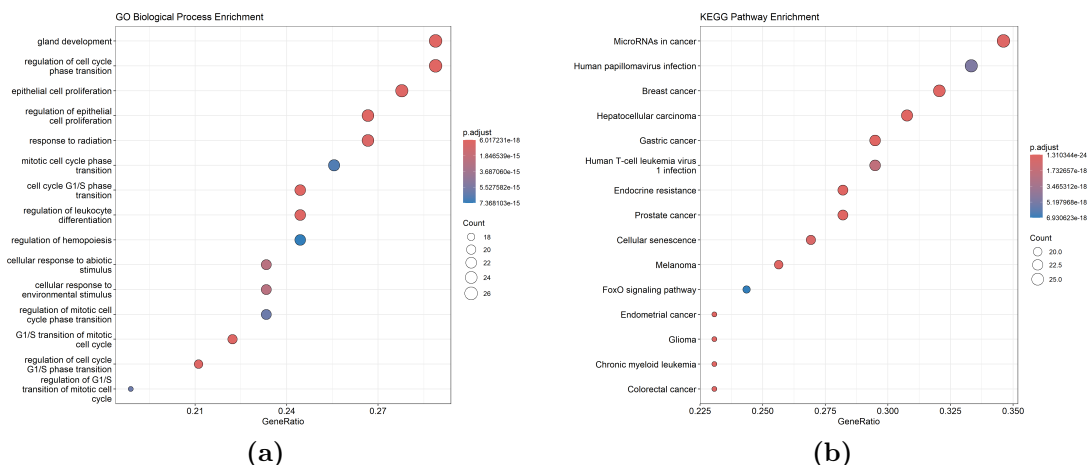


Figure S1: Pathway enrichment analysis of the DRIVER_CORE gene set. (a) Gene Ontology Biological Process enrichment. (b) KEGG pathway enrichment. Dot size indicates the number of genes; colour indicates adjusted p-value.

1.2 KEGG Pathway Enrichment

KEGG pathway analysis identified 68 significantly enriched pathways (adjusted $p < 0.05$). The most significantly enriched pathway was Breast cancer (29 genes; adjusted $p = 1.31 \times 10^{-24}$), followed by Endocrine resistance (22 genes; adjusted $p = 1.79 \times 10^{-24}$) and the PI3K-Akt signalling pathway (27 genes; adjusted $p = 3.63 \times 10^{-16}$). Additional cancer-related pathways with significant enrichment included Proteoglycans in cancer (21 genes; adjusted $p = 1.12 \times 10^{-14}$), Prostate cancer (16 genes; adjusted $p = 6.93 \times 10^{-14}$), and Hepatocellular carcinoma (17 genes; adjusted $p = 3.63 \times 10^{-13}$). Enrichment was also observed for Cellular senescence (16 genes; adjusted $p = 5.96 \times 10^{-12}$), p53 signalling pathway (11 genes; adjusted $p = 3.03 \times 10^{-10}$), and Cell cycle (13 genes; adjusted $p = 1.06 \times 10^{-9}$). Cross-cancer enrichment was evident, with significant results for Melanoma (12 genes; adjusted $p = 1.02 \times 10^{-11}$), Gastric cancer (14 genes; adjusted $p = 1.71 \times 10^{-10}$), and Colorectal cancer (12 genes; adjusted $p = 3.85 \times 10^{-10}$). The top 15 enriched KEGG pathways are shown in Figure S1b, and the complete list is available in SM-KEGG.

2 Breast Cancer Prognosis

2.0.1 In Distribution Experiment Results

Table S1: Benchmarking survival models under the in-distribution (InD) setting.

Dataset	TabSurv_M	DeepHS	DeepSurv	ENCOx	LH	MTLR	PCHazard	PMF	RSF
METABRIC	0.6265	0.5512	0.5923	0.5993	0.5872	0.5868	0.5929	0.5866	<u>0.6002</u>
	± 0.0079	± 0.0191	± 0.0076	± 0.0119	± 0.0218	± 0.0286	± 0.0148	± 0.0088	± 0.0139
NKI	0.6980	0.5588	0.6572	0.6290	0.6522	0.6391	<u>0.6584</u>	0.6268	0.6574
	± 0.0243	± 0.1030	± 0.0601	± 0.0236	± 0.0190	± 0.0341	± 0.0176	± 0.0412	± 0.0469

Notes: LH, Logistic Hazard; PMF, Probability Mass Function; DeepHS, DeepHitSingle; MTLR, Multi-Task Logistic Regression; RSF, Random Survival Forest. Best-performing methods are shown in bold; second-best methods are underlined.

For the in-distribution scenario, each dataset is partitioned using a single random split, with 50% of the samples allocated for training and 50% reserved for testing, using five random seeds (40–44) to ensure reproducibility. We evaluate on the 2 largest cohorts, one per endpoint: METABRIC for relapse-free survival (RFS), and NKI for distant metastasis-free survival (DMFS). Table S1 demonstrates that TabSurv_M achieves the highest C-index on both datasets (0.6265 on METABRIC and 0.6980 on NKI), outperforming all eight baseline survival models. The second-best method varies by dataset: RSF on METABRIC (0.6002) and PCHazard on NKI (0.6584), with TabSurv_M exceeding both by a margin of roughly 2–4 points in C-index.

Across both datasets, all baseline models maintain C-index values comfortably above 0.5, indicating no catastrophic failures on this benchmark; however, DeepHS shows the weakest and least stable performance, with a notably higher standard deviation on NKI (± 0.1030) compared to all other methods, suggesting sensitivity to the specific train/test split and random seed. In contrast, TabSurv_M maintains competitive standard deviations while achieving the top mean C-index on both datasets, potentially benefiting from the robust prior knowledge inherent in the foundation model and the proposed two-stage strategy for handling censored data.

3 Treatment Recommendation Evaluation

3.1 Comparison of TabSurv Recommendation performance across 3 Gene sets

Table S2 summarises the difference in mean survival time between patients who followed versus did not follow the TabSurv treatment recommendation across the three investigated gene sets. The *Nik-94 genes* set demonstrates the largest survival separation, with a mean difference of 1.261 years between the Followed and Not Followed groups. The *Pereira-40 genes* set shows a more modest, separation of 0.693 ± 0.246 years, while the *Mills-45 genes* set exhibits the smallest mean difference (0.469 ± 0.576 years). These

Table S2: Comparison of Delta mean survival time (years) between the *Followed* and *Not Followed* groups across the three gene sets under the TabSurv recommendation framework.

Gene Set	Δ mean survival (mean $\pm$ std)
Nik-94 genes	1.261 $\pm$ 1.992
Mills-45 genes	0.469 $\pm$ 0.576
Pereira-40 genes	0.693 $\pm$ 0.246

Δ Mean survival denotes the difference between *Followed* Recommendation and *Not Followed* Recommendation groups.

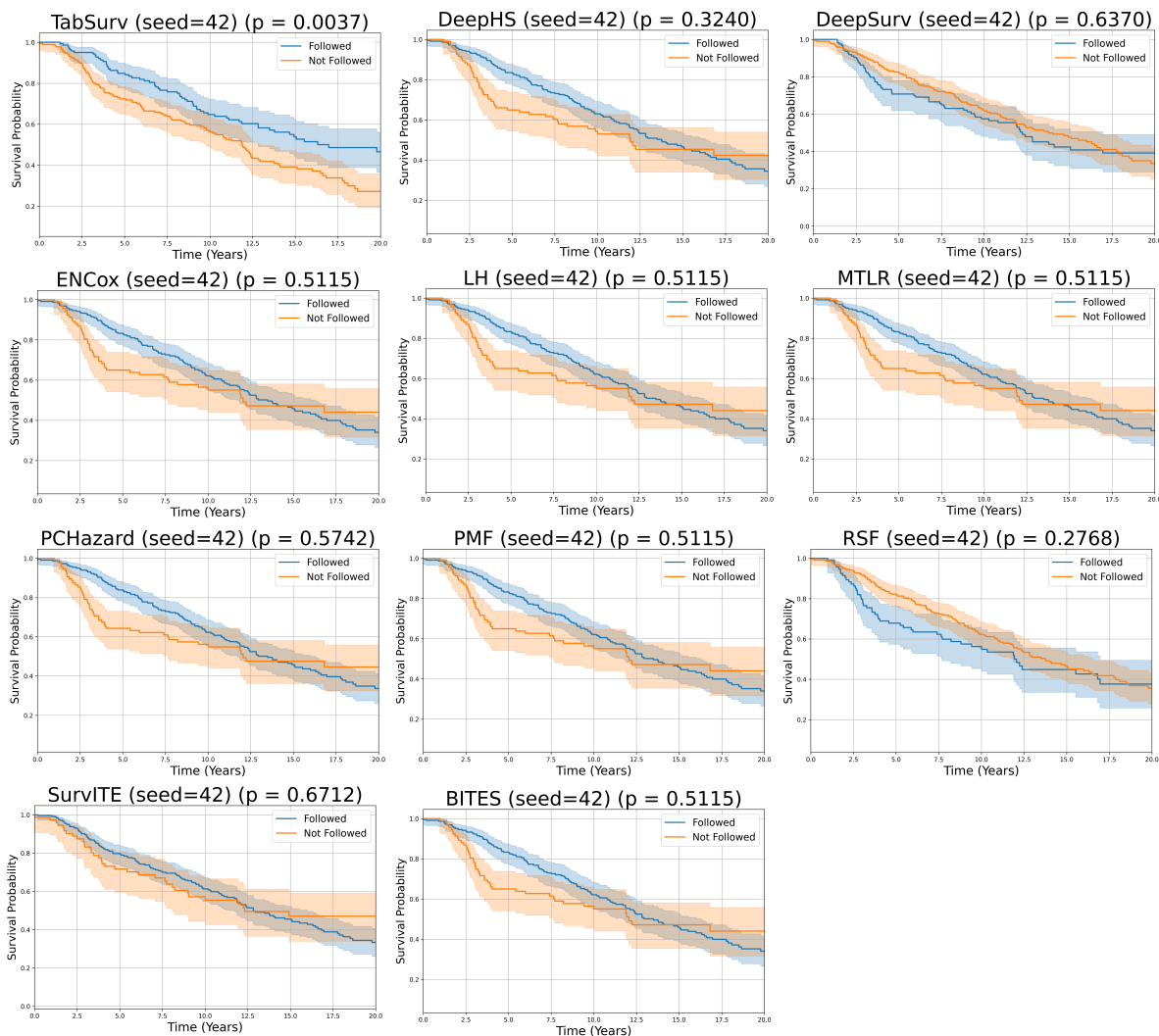


Figure S2: Kaplan-Meier survival curves for personalised treatment recommendation using selected cancer driver genes. For TabSurv and each baseline model, patients are stratified into *Followed* and *Not Followed* groups based on whether the treatment actually received matches the model-recommended therapy. Shaded areas represent 95% confidence intervals. Group differences are assessed using the log-rank test, with corresponding p -values reported.

results indicate that the *Nik-94 genes* DRIVER_CORE set provides the strongest discriminatory signal for treatment recommendation within the TabSurv framework, albeit with wider variability across cohorts than the other two gene sets.

3.2 Kaplan-Meier Curve of TabSurv Recommendation with Baseline Methods

Figure S2 presents Kaplan–Meier survival curves for all baseline methods when using the curated set of biologically important genes, compared to the TabSurv method. The curves show that, for TabSurv, the Followed group consistently exhibits a higher survival probability than the Not Followed group, whereas the baseline methods do not demonstrate a clear long-term survival benefit up to the maximum follow-up time.

References

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- [4] Minoru Kanehisa and Susumu Goto. Kegg: Kyoto encyclopedia of genes and genomes. *Nucleic Acids Research*, 28(1):27–30, 2000.